Bilkent University
Department of Industrial Engineering
IE 468 - Pricing and Revenue Optimization
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DEMAND MODELING and PRICING for Carbonated Drinks

Coca-Cola is analyzing the demand for its carbonated drinks that are offered in retail stores in Turkey. Specifically, the product in consideration is Coke Original. The carbonated drink market is dominated by two companies in Turkey: Coca-Cola and Pepsi. Coca-Cola offers Coke Original, Coke Sugar Free and Sprite, whereas Pepsi offers Pepsi Cola and Pepsi Cola Twist (Cola drink with lime flavor). Migros also sells a private label product (manufactured by a different company) “Migros Cola” and you can assume that this is a base product. Coca-Cola has partnership with Migros, so it was able to collect demand data for 45 days in end of February of 2025 in a particular store which Migros believes is representative of consumer behavior in Turkey. In each of these days, the retail price is recorded as well as the resulting sales. In an effort to understand the effect of price on demand, Coca-Cola changed the wholesale price of Coke considerably which is also reflected in retail price by Migros (The prices of other products may have changed over time too). The collected data can be seen “SalesData” worksheet of the excel file provided. In each of these days, there were, on the average, 165 customers that were interested in carbonated drinks. In each of these days, Coke Original was not out of stock (Hence Sales was equal to Demand).

Question 1: Using the data that you are provided in “SalesData” sheet, fit a linear price-response function that explains how demand of Coke Original changes as a function of its own price (using regression). Explain the results of your regression analysis. Do you think there was a good fit? Explain using the results of your regression analysis.

Question 2: Between Feb 22 and Feb 28, Migros promoted Coke Original by placing the picture and price of the product in Migros’s weekly consumer promotion flier. Using regression again, find how the demand of Coke Original changes as a function of its own price <u>and</u> whether it was placed on weekly promotion. Do you think there was a better fit this time? Explain using the results of your regression analysis.

Question 3: Based on your analysis in Question 2, what do you think the demand for Coke Original will be in this store if the retail price is 42.50 TL and the product is not placed on weekly promotion? How confident are you in your answer? If you were to stock up inventory for daily demand (on a day where you charge 42.50 TL retail price), how many units will you start the day if you want to have a no stock-out probability of 99.5%?

Question 4: Migros always charges 120% of the wholesale price (the price that Coca-Cola charges to Migros) when determining its final retail price. One can of Coke Original costs Coca-Cola 26.50 TL. Using the price-response function in Question 2, find the profit maximizing price for Coca-Cola (when product is not placed on weekly promotion). What is the expected demand in a given day for Coke Original in this Migros store? What is Coca-Cola’s expected daily profit in this store just from Coke Original? (For this part, assume Migros always has enough inventory to satisfy demand and inventory carrying costs are negligible).

Question 5: Now consider the same setting as Question 4 but now assume that Coke Original is placed on weekly promotion. What is the profit maximizing price for Coca-Cola and what is the corresponding daily profit? What is the value of Migros putting Coke Original on promotion for Coca-Cola? (for a day).

Question 6: Now use the data in "SalesData" to fit an exponential price response function and constant elasticity price response function that explain the relationship between the price and demand (for Coke Original) (Also consider the impact of whether the product is placed on weekly promotion). Which price response function better fits the data? What is your metric to compare?

Recognizing the fact that competition may have played a role, Coca-Cola decides to pay attention to not only its own prices, but also to competitors' actions. Fortunately, the consumer choice data for the 45 days of sale in end of February was also available. For each of these days, we know the prices and the sales for each product. This more detailed sales data is provided in "DetailedSalesData" worksheet of the provided spreadsheet. For each day, one of the 6 products may have been out of stock. This is represented in "Stockout?" column (assume the stock out, if any, is for the entire day). Migros also selects one of these products for promotion and advertise them in the weekly flier. The product which is selected for promotion is marked in "Weekly promotion?" column in the file. Assume that the Migros sales market and Migros Cola are managed independently. You do not need to consider whether Migros provides any favour or incurs any loss for its own carbonated drink. Migros Cola is merely a base product available in the market, and its data is available.

Question 7: Based on the data provided in "DetailedSalesData" worksheet, estimate an MNL model for consumer choice for carbonated drink customers. For this purpose, assume that μ=40 and estimate only the average utility of each product attribute (also the extra utility of "weekly promotion" specification).

Question 8: What is the predicted demand for Coke Original for each of the 45 days using the model that you found in Question 7? (Assume that 165 consumers arrive each day and each one of them buys at most one can of carbonated drink). Which price-response model fits the data better? Linear price-response with promotion effect (Question 2) or Multinomial Logit? Also, assume that fractional values of carbonated drink can be sold for each carbonated drink brand so that you can compare with the actual demand data. What metric do you use to justify your claim? Also explain the reasons (qualitatively) for why that model performs better.

Question 9: What is the market share of each product predicted by the model that you found in Question 7. Assume the following prices. Assume that no product is out of stock and no product is selected for "weekly promotion". Comment on the market shares that you found.

Product	Coke Original	Sprite	Pepsi Cola	Coke Sugar Free	Pepsi Cola Twist	Migros Cola
Price	42.50 TL	42.50 TL	38.00 TL	42.00 TL	45.00TL	30.00 TL

Question 10: What is the market share of each product, if Migros increases the prices of all carbonated sodas by 10% from what is given in Question 9?

Question 11: Considering its feasibility issues, Peps, decided to take “Pepsi Twist” off the market. What is the effect of this on the market shares of remaining five products in the market? Assume the following prices:

Product	Coke Original	Sprite	Pepsi Cola	Coke Sugar Free	Migros Cola
Price	42.50 TL	42.50 TL	38.00 TL	42.00 TL	30.00 TL

Comment on the results that you found.

Question 12: After the move by Pepsi, Coca-Cola found an opportunity to increase its market share further and decided to introduce one more product to the market: Coke Twist. Assuming that the product utilities are linear in its attributes, calculate the market share of each product. What percent of Pepsi customers in Question 10 now switches to Coca-Cola? Assume that the following retail prices will be in effect:

Product	Coke Original	Sprite	Pepsi Cola	Coke Sugar Free	Coke Twist	Migros Cola
Price	42.50 TL	42.50 TL	38.00 TL	42.00 TL	46.00TL	30.00 TL

Question 13: Now consider the same setting as in Question 11, but now Sprite is selected for “weekly promotion” by Migros. Assume no product is out of stock. Find the market share of each product using the MNL model you estimated in Question 6.

Question 14: Going back to the original setting, assume none of the products are stockout or are on weekly promotion. For the 46th day you happen to find the price data as follows:

Product	Coke Original	Sprite	Pepsi Cola	Coke Sugar Free	Pepsi Cola Twist	Migros Cola
Price	41.50 TL	41.00 TL	37.20 TL	42.0 TL	45.50 TL	31.00 TL

Assume that 165 consumers arrive on day 46th and each one of them buys at most one can of a carbonated drink, what would be your expected demand for each product. You can round your answers to the nearest integer. Explain how you obtained the demand data.